

Dalla lezione precedente

(X, Y) v.c. congiuntamente continue

$$f(x, y) = \begin{cases} \alpha(x+y+1) & \text{se } x, y \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

$$\alpha = \frac{1}{2} \quad (\text{si verifica che } 1 = \iint_{\mathbb{R}^2} f(x, y) dx dy)$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$$f_X(x) = \begin{cases} \frac{1}{2}(x + \frac{3}{2}) & \text{se } x \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}(y + \frac{3}{2}) & \text{se } y \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

$$\begin{aligned}
 E[X] &= \int_{-\infty}^{+\infty} x f(x) dx = \int_0^1 x \left(\frac{1}{2} \left(x + \frac{3}{2} \right) \right) dx = \\
 &= \int_0^1 \left(\frac{1}{2} x^2 + \frac{3}{4} x \right) dx = \left[\frac{x^3}{6} + \frac{3}{8} x^2 \right]_0^1 = \frac{1}{6} + \frac{3}{8} = \\
 &= \frac{4 + 9}{24} = \frac{13}{24}
 \end{aligned}$$

$$\begin{aligned}
 E[X^2] &= \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^1 x^2 \left(\frac{1}{2} \left(x + \frac{3}{2} \right) \right) dx = \\
 &= \int_0^1 \left(\frac{x^3}{2} + \frac{3}{4} x^2 \right) dx = \left[\frac{x^4}{8} + \frac{x^3}{4} \right]_0^1 = \\
 &= \frac{1}{8} + \frac{1}{4} = \frac{3}{8}
 \end{aligned}$$

$$\text{Var}(X) = \frac{3}{8} - \left(\frac{13}{24} \right)^2 = \frac{216 - 169}{24^2} = \frac{47}{24^2}$$

$$E[XY] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \underbrace{xy}_{\text{...}} f(x,y) dx dy =$$

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{xy}_{h(x,y)} f(x,y) dx dy =$$

$$= \int_0^1 \int_0^1 xy \frac{1}{2} (x+y+1) dx dy = \frac{1}{2} \int_0^1 y \left(\int_0^1 (x^2 + xy + x) dx \right) dy =$$

$$= \frac{1}{2} \int_0^1 y \left[\frac{x^3}{3} + y \frac{x^2}{2} + \frac{x^2}{2} \right]_0^1 dy =$$

$$= \frac{1}{2} \int_0^1 y \left(\underbrace{\frac{1}{3} + \frac{1}{2}}_{\frac{2+3-5}{6}} + \frac{1}{2} y \right) dy = \frac{1}{2} \int_0^1 \left(\frac{5}{6} y + \frac{1}{2} y^2 \right) dy =$$

$$= \frac{1}{2} \left[\frac{5}{6} \frac{y^2}{2} + \frac{1}{6} y^3 \right]_0^1 = \frac{1}{2} \left(\frac{5}{12} + \frac{1}{6} \right) = \frac{5+2}{12 \cdot 2} = \frac{7}{24}$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] = \frac{7}{24} - \left(\frac{13}{24}\right)^2 \\ &= \frac{168 - 169}{24^2} = -\frac{1}{24^2} \end{aligned}$$

$$= \frac{168 - 162}{24^2} = -\frac{1}{24^2}$$

$$\text{Corr}(X, Y) = \frac{-\frac{1}{24^2}}{\sqrt{\frac{47}{24^2} \cdot \frac{47}{24^2}}} = -\frac{1}{24^2} \cdot \frac{24^2}{47} = -\frac{1}{47}$$

8.2 $Z = \min(X, Y)$ $F_Z(a)$?

$$\begin{aligned} F_Z(a) &= P(Z \leq a) = P(\min(X, Y) \leq a) = \\ &= 1 - P(\min(X, Y) > a) = 1 - P(X > a, Y > a) = \\ &= 1 - \int_a^{+\infty} \int_a^{+\infty} f(x, y) dx dy = 1 - \begin{cases} 0 & \text{se } a > 1 \\ \int_a^1 \int_a^1 \frac{1}{2}(x+y+1) dx dy & \text{se } 0 \leq a \leq 1 \\ 1 & \text{se } a < 0 \end{cases} \end{aligned}$$

$$\frac{1}{2} \int_a^1 \left(\int_a^1 (x+y+1) dx \right) dy = \frac{1}{2} \int_a^1 \left[\frac{x^2}{2} + yx + x \right]_a^1 dy =$$

1 2 7.

$$\begin{aligned}
&= \frac{1}{2} \int_a^1 \left[\frac{1}{2} + y + 1 - \frac{a^2}{2} - ay - a \right] dy = \\
&= \frac{1}{2} \left[\frac{3}{2}y + \frac{y^2}{2} - \frac{a^2}{2}y - ay - \frac{ay^2}{2} \right]_a^1 = \\
&= \frac{1}{2} \left[\frac{3}{2} + \frac{1}{2} - \frac{a^2}{2} - a - \frac{a}{2} - \frac{3}{2}a - \frac{1}{2}a^2 + \frac{a^3}{2} + \frac{a^2}{2} + \frac{a^3}{2} \right] = \\
&= \frac{1}{2} (a^3 - 3a + 2)
\end{aligned}$$

$$F_2(a) = \begin{cases} 1 & \text{se } a > 1 \\ 1 - \frac{1}{2}(a^3 - 3a + 2) & \text{se } a \in [0, 1] \\ 0 & \text{se } a < 0 \end{cases}$$

es. X v.c. cont. n.r. con $f(x) = \begin{cases} kx & \text{se } x \in (1, 2] \\ 0 & \text{altrove} \end{cases}$

$$k! \quad Y = \sqrt{X} \quad f_Y(y), F_Y(y)? \in [Y]$$

$$X_1, \dots, X_{100}$$

..., ..., 100

$$P(\bar{X} > 1)?$$

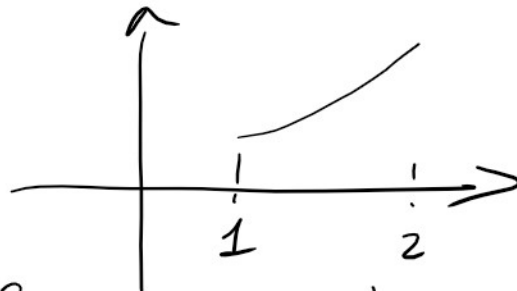
$$f(x) = \begin{cases} kx & x \in [1, 2] \\ 0 & \text{altrimenti} \end{cases}$$

$$f(x) \geq 0 \quad \forall x \in \mathbb{R} \Rightarrow k \geq 0$$

$$1 = \int_{-\infty}^{+\infty} f(x) dx = \int_1^2 kx dx = k \left[\frac{x^2}{2} \right]_1^2 = k \left(\frac{4}{2} - \frac{1}{2} \right) = \frac{3}{2} k$$

$$\Rightarrow k = \frac{2}{3}$$

$$\text{Se } Y = \sqrt{X} = g(x)$$



Dato che $\sqrt{x}$ è una funzione monotona crescente in $[1, 2]$

$$f_Y(a) = f_X(g^{-1}(a)) \frac{d g^{-1}(a)}{da}$$

$$\begin{aligned}
 \sqrt{x} = a &\Rightarrow x = a^2 = g^{-1}(a) \\
 \frac{d g^{-1}(a)}{da} &= 2a; \quad f_X(g^{-1}(a)) = \begin{cases} \frac{2}{3} g^{-1}(a) & \text{se } g^{-1}(a) \in [1, 2] \\ 0 & \text{otherwise} \end{cases} \\
 &= \begin{cases} \frac{2}{3} a^2 & \text{se } a^2 \in [1, 2] \\ 0 & \text{otherwise} \end{cases} \\
 &= \begin{cases} \frac{2}{3} a^2 & \text{se } a \in [1, \sqrt{2}] \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

$$f_Y(a) = \begin{cases} \frac{2}{3} a^3 & \text{se } a \in [1, \sqrt{2}] \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 F_Y(a) &= \begin{cases} 0 & \text{se } a < 1 \\ \int_1^a \frac{2}{3} y^3 dy & \text{se } a \in [1, \sqrt{2}] \\ 1 & \text{se } a > \sqrt{2} \end{cases} \\
 &= \begin{cases} 0 & \text{se } a < 1 \\ \left[\frac{1}{3} y^4 \right]_1^a = \frac{a^4 - 1}{3} & \text{se } a \in [1, \sqrt{2}] \\ 1 & \text{se } a > \sqrt{2} \end{cases}
 \end{aligned}$$

$$E[Y] = \int_{-\infty}^{+\infty} y f_Y(y) dy = \int_1^{\sqrt{2}} y \frac{4}{3} y^3 dy = \left[\frac{4}{3} \frac{y^5}{5} \right]_1^{\sqrt{2}} =$$

$$= \frac{4}{3} \left[\frac{2^{5/2} - 1}{5} \right]$$

$$\mu = E[X_k], \sigma^2 = \text{Var}(X_k)$$

$X_1, \dots, X_{100}$ i.i.d.

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{100}}\right)$$

$$P(\bar{X} > 4) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{100}}} > \frac{4 - \mu}{\frac{\sigma}{\sqrt{100}}}\right) =$$

$$= P\left(Z > \frac{4 - \mu}{\frac{\sigma}{\sqrt{100}}}\right) = 1 - \Phi\left(\frac{4 - \mu}{\frac{\sigma}{\sqrt{100}}}\right)$$

$$\mu = E[X_k] = \int_{-\infty}^{+\infty} x f(x) dx = \int_1^2 \frac{2}{3} x \cdot x dx =$$

$$= \frac{2}{3} \left[\frac{x^3}{3} \right]_1^2 = \frac{2}{9} (8 - 1) = \frac{14}{9}$$

$$E[X_k^2] = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_1^2 x^2 \frac{2}{3} x dx =$$

$$E[X_K^2] = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_1^2 x^2 \frac{2}{3} x dx =$$

$$= \frac{2}{3} \left[\frac{x^4}{4} \right]_1^2 = \frac{2}{3} \left(\frac{16-1}{4} \right) = \frac{5}{2}$$

$$\text{Var}(X_K) = \frac{5}{2} - \left(\frac{14}{9} \right)^2 = \frac{405 - 2 \times 14^2}{162} = \frac{405 - 2 \times 196}{162} = \frac{13}{162}$$

$$\bar{X} \sim N\left(\frac{14}{9}, \frac{1}{10} \sqrt{\frac{13}{162}}\right)$$

es. $X_1, X_2, X_3, X_4 \sim N(1, 1)$ INDIP.

$$P(X_1 + X_2 = X_3 + X_4) = 0$$

$$P(4X_1 > 1) = P(X_1 + X_2 + X_3 + X_4 > 1)$$

in altre parole $X_1 + X_2 + X_3 + X_4$ è
equivalente a $4X_1$?

le X_K possono avere valori diversi
(es. lancio di 4 dadi) mentre $4X_1$

le X_k possono avere valori diversi
(es. lancio di 4 dadi) mentre $4X_1$
implicz che $X_2 = X_3 = X_4 = X_1$

$$P(X_1 + X_2 + X_3 + X_4 > 1)$$

per riproducibilit  $Y = \sum_{k=1}^4 X_k \sim N$

$$E[Y] = E\left[\sum_{k=1}^4 X_k\right] = \sum_{k=1}^4 E[X_k] = 4$$

$$\text{Var}(Y) = \text{Var}\left(\sum_{k=1}^4 X_k\right) = \sum_{k=1}^4 \text{Var}(X_k) = 4$$

$$Y \sim N(4, 2)$$

$$\begin{aligned} P(Y > 1) &= P\left(\frac{Y-4}{2} > \frac{1-4}{2}\right) = P(Z > -\frac{3}{2}) = \\ &= 1 - \Phi_2\left(\frac{3}{2}\right) = 1 - \Phi_2(1.5) \end{aligned}$$

$$P(X_1 + X_2 > X_3 + X_4) = P(X_1 + X_2 - X_3 - X_4 > 0)$$

$$S = X_1 + X_2 - X_3 - X_4 \sim N(?, ?)$$

$$E[S] = E[X_1] + E[X_2] - E[X_3] - E[X_4] = 0$$

$$\text{Var}(S) = \text{Var}(X_1) + \text{Var}(X_2) + (-1)^2 \text{Var}(X_3) + (-1)^2 \text{Var}(X_4) = 4$$

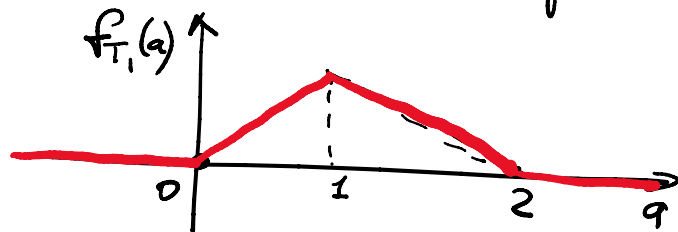
$$S \sim N(0, 2)$$

$$P(S > 0) = P\left(\frac{S-0}{2} > \frac{0-0}{2}\right) = P(Z > 0) = 1 - F_Z(0) = \frac{1}{2}$$

es. $X_1, X_2, X_3, X_4 \sim U(0,1)$ INDIP.

$$P(X_1 + X_2 > X_3 + X_4) ?$$

Abbiamo già visto che $T_1 = X_1 + X_2$ ha
funzione di densità di prob.



$$f_{T_1}(a) = \begin{cases} a & \text{se } a \in [0,1] \\ 2-a & \text{se } a \in [1,2] \\ 0 & \text{altrimenti} \end{cases}$$

$T_2 = X_3 + X_4$ presenterà la stessa $f_{T_2}(a) = f_{T_1}(a)$

$$P\left(\lim_{n \rightarrow \infty} T_1 > \lim_{n \rightarrow \infty} T_2\right) = \quad \text{con } T_1 \text{ e } T_2 \text{ INDIPENDENTI}$$

$$P(t_1 > t_2) = \text{con } t_1 \text{ e } t_2 \text{ INDIPENDENTI}$$

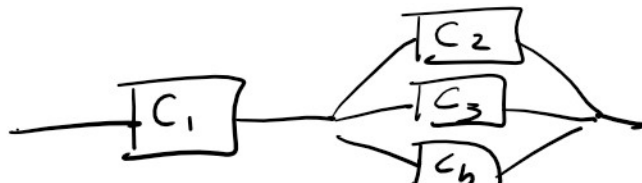
$$= \int_{-\infty}^{+\infty} \left(\int_{t_2}^{+\infty} f_{T_1}(t_1) f_{T_2}(t_2) dt_1 \right) dt_2 =$$

$$= \int_0^2 \left(\int_{t_2}^2 f_{T_1}(t_1) f_{T_2}(t_2) dt_1 \right) dt_2 =$$

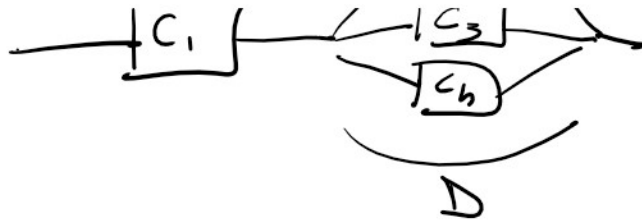
$$= \int_0^1 f_{T_2}(t_2) \left(\int_{t_2}^2 f_{T_1}(t_1) dt_1 \right) dt_2 + \int_1^2 f_{T_2}(t_2) \left(\int_{t_2}^2 f_{T_1}(t_1) dt_1 \right) dt_2 =$$

$$= \int_0^1 t_2 \left(\int_{t_2}^1 1 dt_1 + \int_1^2 (2-t_1) dt_1 \right) dt_2 +$$

$$+ \int_1^2 (2-t_2) \left(\int_{t_2}^2 (2-t_1) dt_1 \right) dt_2$$



$X_k = \text{durata di}$
 funzionamento
 di D_k



funzionamento
di D_k

Y = durata di funzionamento
di D

T = 'durata di funzionamento del sistema'

$$T = \min(X_1, Y) \quad \text{con } Y = \max(X_2, X_3, X_4)$$

$$\begin{aligned} F_T(a) &= P(\min(X_1, Y) \leq a) = \\ &= 1 - P(\min(X_1, Y) > a) = 1 - P(X_1 > a) P(Y > a) = \\ &= 1 - (1 - F_{X_1}(a)) (1 - \underbrace{P(Y \leq a)}) \end{aligned}$$

$$\begin{aligned} F_Y(a) &= P(Y \leq a) = P(\max(X_2, X_3, X_4) \leq a) = \\ &= P(X_2 \leq a, X_3 \leq a, X_4 \leq a) = \\ &= P(X_2 \leq a) P(X_3 \leq a) P(X_4 \leq a) = F_{X_2}(a) F_{X_3}(a) F_{X_4}(a) \end{aligned}$$

$$\Rightarrow \underbrace{F_T(a)} = 1 - (1 - F_{X_1}(a)) (1 - F_{X_2}(a) F_{X_3}(a) F_{X_4}(a))$$

$$P(T > 180) = 1 - F_T(180) = (1 - F_{X_1}(180))(1 - F_{X_2}(180))F_{X_3}(180)F_{X_4}(180)$$

Q5 settembre 2018

$$Y \in \{3, 4, 5\}$$

$$P_G + P_L = 1$$

$$P_L = 2P_G \Rightarrow P_G = \frac{1}{3}, P_L = \frac{2}{3}$$

$$\begin{aligned} P(Y=3) &= P(G_1 \cap G_2 \cap G_3 \cup L_1 \cap L_2 \cap L_3) = \\ &= \left(\frac{1}{3}\right)^3 + \left(\frac{2}{3}\right)^3 \end{aligned}$$

$$\begin{aligned} P(Y=4) &= P((L_1 \cap G_2 \cap G_3 \cap G_4) \cup (G_1 \cap L_2 \cap G_3 \cap G_4) \cup \\ &\quad (G_1 \cap G_2 \cap L_3 \cap G_4) \cup (G_1 \cap L_2 \cap L_3 \cap L_4) \cup \\ &\quad (L_1 \cap L_2 \cap G_3 \cap L_4) \cup (L_1 \cap G_2 \cap L_3 \cap L_4) \cup \\ &\quad (G_1 \cap G_2 \cap G_3 \cap L_4) \cup (L_1 \cap L_2 \cap L_3 \cap G_4)) \end{aligned}$$

$$= \frac{2}{3} \cdot \left(\frac{1}{3}\right)^3 \cdot 3 + \frac{1}{3} \cdot \left(\frac{2}{3}\right)^3 \cdot 3$$

$$P(Y=5) = 1 - P(Y=3) - P(Y=4)$$